

**1. Efficient energy conversion** – There is no dearth of energy. It is all around us.

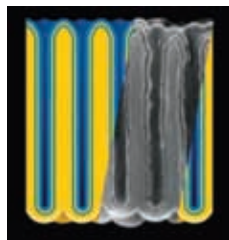
But it is often inaccessible or in a unusable form. Currently, we extract chemical energy from coal, oil and natural gas because these energy-rich materials come in convenient forms. These are also relatively easy to transform into a range of products, including formulated fuels, and into electricity that can be widely distributed by power lines. As fossil fuels grow scarce and increase in value, nanotechnology could be used to reduce losses during energy conversion. For instance, nano-engineered catalysts could improve the conversion of crude oil into various



Solar Panels

petroleum products, as well as the conversion of coal into clean fuels for generating electricity. Over the long term, the attention is likely to shift to the improvement of our ability to convert sunlight into electricity, the easiest form of energy to use. Common commercially available solar cells have an efficiency of about 12 percent; some laboratory models achieve 30 percent. Researchers are testing many different ways to boost energy conversion by fine-tuning the material properties in solar cells, and it is quite likely that the problem will be solved using nanotechnology. Some prototypes have embedded carbon nanotubes in them, while others take advantage of nanocrystals or clusters of atoms called quantum dots. However, we still need to learn a lot more about the fundamental processes behind various energy conversions, including sunlight to electricity, heat gradients to electricity and nuclear fusion to electricity.

- 2. Efficient energy storage** – The next step after efficiently converting energy into a usable form, is to store it efficiently so that we can use it whenever we need it. In the near term, nanotechnology could be used to create appliances and other products that can store energy more efficiently—that is, take up charge and hold it over time. Many research groups are working on better batteries, often using engineered nanomaterials. Better batteries might make it possible to store energy that is generated in a widely distributed way, such as within consumer products. That would eliminate the need to ship energy across thousands of miles and solve the



Energy Storage